

Enhancing Digital Planner Design: Insights from Undergraduate Use of Physical and Digital Planners for Academic Tasks.

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This study aims to optimize academic task planning for college students by examining the design goals and features of both physical and digital planners. Despite technological advancements in digital planning tools, a notable preference for physical planners persists among university students. Through surveys (N = 43), follow-up interviews (N = 10), and observations (N = 10), we delve into the reasons behind this preference, exploring the emotional and practical benefits of physical planners and identifying perceived shortcomings in digital alternatives. Our analysis sheds light on how both planner types impact academic task management, goal fulfillment, emotional well-being, and overall productivity. Our findings suggest avenues for enhancing digital planner interfaces, potentially improving not only academic performance but also mental well-being, self-fulfillment, work-life balance, and personal organization. This research contributes to the field of Human-Computer Interaction by offering insights into user preferences and the nuanced interplay between digital and physical planning tools in contemporary academic settings.

Keywords and Phrases: Planner, Academic Productivity, Task Management, Activity Centered Design

1 INTRODUCTION

In the field of Human-Computer Interaction (HCI), understanding how individuals interact with both digital and physical tools for managing daily tasks offers a valuable area of study. However, research specifically focusing on the usage of planners, particularly for academic tasks, remains limited. Our study aims to fill this gap by examining undergraduate preferences for various planning methods, especially in the context of the recent COVID-19 pandemic, which has dramatically shifted the landscape of work and study towards more remote and digital formats [6]. This shift underscores the vital role of planners in providing structure amid changing workflows and educational environments.

We define a planner as any interface used for inputting and tracking tasks for scheduling or task management. Planners are crucial in daily life, boosting motivation, increasing efficiency, and aiding in organizational planning. They are particularly beneficial for college students who often face rapidly changing schedules and a mix of structured and irregular academic and extracurricular commitments.

Our research is motivated by the diverse range of tools and systems utilized to enhance productivity, spanning digital, physical, and hybrid realms. We aim to understand the underpinnings of planner preferences, the frequency and effectiveness of planner interactions, and the key features that influence user engagement and productivity in an academic context.

Specifically, our study investigates why some university students continue to prefer physical planners, despite the conveniences of digital alternatives, and what unique benefits physical planners provide. Through a mixed-methods approach comprising surveys, interviews, and structured observations, we explore how different planner interfaces impact academic stress management, focus on coursework, and the fulfillment of broader life and career goals. We also aim to identify specific pain points in planner usage, such as the challenges in recording activities, managing tasks from various academic fields, and aligning actual plans with higher-level academic objectives.

The insights from this research contribute to HCI by revealing how user interface design can better align with the motivations and preferences of users. We seek to bridge the gap between user needs and UI design, aspiring to create more effective, user-friendly, and emotionally resonant digital planning interfaces. Our findings have the potential to enhance academic productivity and engagement, making it easier for university students to navigate their academic responsibilities. Additionally, the insights gained from this study could extend to improving digital planners for broader applications, including work-life balance, social planning, and personal organization.

2 RELATED WORK

In the post-pandemic era, characterized by a significant shift towards digital formats for tasks and assignments, there has been a marked acceleration in the adoption of digital planning systems[6]. Yet, research reveals that a substantial number of individuals still prefer physical planning systems, like paper planners and bullet journals[3, 8]. These traditional methods are particularly favored for self-tracking, known for supporting the emotional aspects of planning and tracking in everyday life[3]. This preference is underscored by their greater perceived customizability, reflexivity, and personal significance compared to digital tools[3]. Despite the rise in digital tool usage, non-digital forms of manual tracking and planning continue to hold prominence, attributed to their tangible nature and functional benefits[6]. This preference is also supported by behavioral and neuroscience research, which demonstrates notable differences in brain activation when using paper versus electronic devices. For instance, paper-based note-taking has been associated with deeper memory encoding compared to using tablets[8]. However, existing studies primarily focus on the emotional and well-being aspects of physical planning, such as self-tracking, creative expression, reflection, and mindfulness[2, 3, 6]. There remains a significant research gap in exploring the productivity potential of planners, particularly in academic settings. Our study seeks to bridge this gap, aiming to understand how digital planners can be optimized to enhance academic productivity, while still acknowledging its roles in supporting emotional and functional needs.

2.1 Advantages of Physical Tools

Several researchers have investigated the benefits of physical tools and practices adjacent to planners. Ayobi et. al. investigate the practice of bullet journaling as an approach to self-tracking in everyday life. The study seeks to determine the extent to which it may be superior to digital self-tracking technologies

and how it can offer insights for the future design of self-tracking interfaces [3]. The research explores how individuals adapt their journaling systems to meet practical and emotional needs through a comprehensive analysis of photos and related conversations on social media platforms like Instagram. While the previous work emphasizes customization and reflection, our idea places a stronger emphasis on productivity. Nonetheless, the findings are significant for our study as they offer valuable insights into the design objectives of planners, particularly in emotional aspects like enabling flexibility for creating visualizations that make the tracking process personally meaningful. Additionally, the ability to share or post these visualizations online to foster a sense of community is an important consideration.

Parastoo et. al. further expand work on self-tracking practices, particularly non-digital manual self-tracking methods, with the aim of comprehending users' motivations, challenges, and perceptions related to both physical and digital tracking tools [6]. Survey and follow up interviews were conducted to gather data. The study concludes with design implications for future technologies, including ways in which digital tools can extend and support reflective self-tracking practices. In relation to our idea, both the prior work and our idea center around understanding user motivations, preferences, and interactions with tools—both physical and digital. However, our idea expands the scope by concentrating on planner usage among college students (both physical and digital) rather than self-tracking practices.

Some researchers have found advantages of physical tools compared to digital ones. Huang et. al. investigate the impact of using paper versus mobile calendars on everyday planning and plan fulfillment behavior [11]. It shows that the shift from paper to mobile calendars, driven by convenience, may potentially result in less effective planning and plan implementation. The study shows that this higher fulfillment rate in paper calendar users is attributed to their adoption of a broader, more macroscopic perspective during planning. Additionally, the research suggests that mobile calendar users can enhance their plan quality by adopting this comprehensive perspective. This study aligns with our goal of optimizing planners for task management. Our research could expand on this by investigating UI design elements and features that bridge the gap between paper and mobile planning, enhancing user engagement and productivity.

Keita et. al. find similar benefits to physical tools while examining how the use of paper notebooks versus mobile devices affects memory encoding and retrieval, focusing on the neuroscience aspect [4]. They found that using paper notebooks led to shorter recording times and higher accuracy on simpler tasks, compared to digital devices. Additionally, there was increased brain activation during memory retrieval with paper notebooks, suggesting enhanced memory encoding and retrieval. This implies that the tangible qualities of paper may improve memory performance. Our research, while acknowledging these findings, shifts the focus to academic task planning and productivity among college students, employing both physical and digital planners. We aim to explore how the features of these mediums can be optimized for academic productivity, building upon the cognitive insights from the referenced study. Still, it is important to be aware of such advantages to physical interfaces when understanding user efficacy and designing new interfaces.

Unlike these studies, Velikova et. al. found an advantage of a digital interface over a physical one [9]. In this study, paper and computer touch-screen questionnaire interfaces were tested and evaluated for their effectiveness at collecting data on patients' health and quality of life. Overall, between the paper and touchscreen, the data maintained its quality between the different modes of collection and a notable percentage of the patients preferred the touchscreen over the paper. This work is applicable to our study in the nuances it observes between the different digital and physical interfaces. Like the patients in this study, students may also have physical and digital interface specific usability issues, such as misreading of text on paper, or misentry of information and misclicking, that would affect the effectiveness of their academic planning system.

2.2 Digital Tools

Digital tools have exposed important concepts relevant to our study. To improve student time management, organization, and efficiency in light of recent declines, Mohamad and Azizan design an online student academic planner called UniPlanner using the waterfall method, consisting of the steps requirements, design, implementation, verification, and maintenance [5]. They identify key gaps in individual commercial planner systems - these gaps include lack of distinct reminder notifications, lack of planning functionality, unnecessary manual input, inability to modify previous tasks, and uninformative list view. Our work will not be limited to 4 commercial systems, will address physical planners, and will consider planning systems as a whole in order to uncover key design principles. It will also more thoroughly investigate student preferences.

Planners extend beyond the daily or weekly tasks typically included in college student planners. Zhen et. al. focus on an interactive course planner which supports strategic planning for students to explore and simulate various routes through a curriculum. The aim of this interface is for students to easily navigate the complex curriculum system and find their optimal pathways as they plan and keep track of their route through the curriculum. This study relates to our idea as it builds insights into the design structure of the planning system and how it helps the user to achieve their tasks. With the insights we have gained from this study, our study can further explore how the design elements and features of the planner enhance the productivity of the tasks.

William et. al. explore a planner integrated into a file manager [10]. They investigate the ways to help users in planning and organizing their projects. The paper introduces the Personal Project Planner, which functions as an extension to the file manager and helps to organize project-related information. It also shows how the planner manages different forms of information and tasks, making it easier for the user to keep track of their tasks and those that need to be completed. This study relates to our idea as it builds insights into the planning system and task management that optimizes the task planning experiences and enhances productivity. With the insights we have gained from this study, our study can further explore how the planning system can be designed to effectively manage the tasks and organize the complex information so that the users can easily keep track of all the information.

Several non-planner tools generate insights applicable to digital planners. Amib et. al. examine the Trackly app's support for multiple sclerosis self-management [12]. Trackly, a prototype app, allows users to create and color personalized pictorial trackers. Participants in the study valued the app's flexibility and the ability to color in their trackers, emphasizing the importance of customization in digital health tools. This aligns with our aim to enhance engagement and the overall experience in digital planning systems by improving the features of digital apps and addressing their shortcomings.

Some of these tools use novel interfaces that can be transferred to digital planners. Robertson et. al. address the issue of users spending time arranging and switching between windows by using iterative design to design Scalable Fabric [7]. Scalable Fabric, a windows manager in which windows outside of the focus view remain visible in shrunken form around the periphery, was preferred by users to the legacy window manager. This work is transferable to planners, providing insight into how to maintain an unobtrusive planner presence. Our work will incorporate this idea when considering how to more broadly support college student preferences for academic tasks.

3 SURVEY

We developed a survey comprising both quantitative and qualitative questions to understand the use of planners for academic tasks. Participants were categorized based on their use of a physical planner, digital planner, or a combination of both. The survey included Likert scale questions to assess user attitudes toward their planning system and its efficacy. Additionally, open-ended questions explored motivations, preferred features, and challenges faced by the users. Open-ended questions can be found in the Appendix (section A).

3.1 Recruitment

The survey was conducted online and remained open for approximately one week. We primarily used snowball sampling to gather responses, distributing the survey to undergraduate students through various social media platforms such as Discord, Instagram, and Gmail. While snowball sampling was effective in enhancing our data collection, we acknowledge the sampling bias, homogeneity bias and other potential biases this method may introduce, as it relies on us and our respondents sharing the survey within our networks.

4 SURVEY RESULTS

4.1 Demographics

Overall, 43 people responded to our survey. The demographics of our participants consisted primarily of undergraduate students at UCSD between the ages of 18-24, and was largely composed of seniors and juniors (34 out of 43 total). We surveyed a wide variety of majors (>30 distinct majors) with a slight bias for Cognitive Science students (5) and a strong bias for STEM majors (28 out of 43).

4.2 Quantitative Survey Question Results

We found that the majority of respondents were comfortable with using digital tools, with 79.1% of them rating at a 4 or 5 on the Likert scale (1 - low level of comfort, 5 - high level of comfort). In terms of planner reliance, respondents were less dramatically distributed, with 58.1% of respondents (25 out of 43) rating their reliance at a 4 or 5 (1 = not reliant at all, 5 = very reliant) and 25.6% of respondents (11 out of 43) rating at a 1 or 2.

Out of the 3 options (Digital, Physical, and Combination of physical and digital), physical planning systems were least popular at 15.6%, compared to 39.5% of users who use Digital planning systems and 44.2% of users who use Combination planning systems. The top 3 planning tools were Google Calendar, Apple Notes, and Canvas Calendar. Google Calendar was dramatically more popular, with over 34% of respondents citing it among their tools, compared to 11.6% of respondents citing Apple Notes. When each group rated the freedom, clarity, ease of use and execution, and learning curve of their planners from 1-5, physical users on average rated the freedom of their planning systems more highly than digital or combo users (averages: digital - 3.9, combo - 3.7, physical - 4.9). Physical users also rated their systems slightly better with clarity and learning curve, but ease of use and execution was about the same across all 3 groups. For accomplishing different goals, we evaluated the effectiveness of the physical and digital planners separately. On average, physical planners were rated higher on organizing long-term projects and tracking deadlines, while digital planners were rated higher with daily task management. Digital and physical planners were rated the same on managing work-life balance. On rating planner features by importance, participants across all 3 groups of users on average rated the ease of use of the planner the highest, and integration with other tools as the lowest.

4.3 Qualitative Survey Question Results

We asked 4 long answer questions of all respondents , and one extra long answer question of combined respondents.

4.3.1 Factors Influencing Planner Choice

Accessibility/Convenience was the most frequently mentioned theme by both Combined and Digital respondents, while Customizability and Previous Experience/Habit were most commonly mentioned by physical respondents.

4.3.2 Aspects Appreciated

Users across categories most frequently mentioned Ease of Use. Digital and combined respondents noted Accessibility/Convenience, but physical respondents did not.

4.3.3 Problems Encountered

Combined respondents were the only ones to mention Difficulty Planning, Lack of Presence, Unclear/Difficult Controls, Difficulty Modifying, and Inaccessibility. Digital respondents most frequently noted Uninformative View, and physical respondents most frequently noted User Habits/Consequences, a theme encompassing responses that mention user actions.

4.3.4 Aspects Missing/Improvements

Responses were varied, but Reminders were frequently mentioned and Digital users often noted Synchronization. Many physical respondents cited user behaviors or faulty memory rather than design elements.

4.3.5 Combined User Preferences

Participants who mentioned short or long term tasks uniformly preferred a physical planner for short-term or day-to-day tasks and a digital planner for long-term or large tasks. Similarly, physical planners were preferred for initially writing tasks down, while digital planners were preferred for tracking or viewing tasks.

5 FOLLOW-UP INTERVIEWS AND OBSERVATIONS

After conducting a survey, we followed up with participants interested in more in-depth interviews to discuss their planner usage and planning processes. We also conducted observations, assigning participants a planning task and recording how they utilized their current planning systems to complete it. We obtained responses from a total of 10 participants, who were both interviewed and observed. Our selection criteria were based solely on availability, without considering other demographic factors. The interview questions, extending the survey queries, delved deeper into qualitative aspects, exploring detailed planning experiences, emotional factors, outcomes etc. Each interview lasted approximately 25 minutes, while the duration of structured observations varied, depending on each participant's planning habits. Sample interview questions and details of the observation can be found in Appendix (section B).

5.1 Demographics

Our research team conducted interviews and observations with 10 college students who responded to our survey, with 7 identified as females and 3 identified as nonbinary. These students ranged from sophomores to seniors, aged between 19 and 24 years, with the majority falling within the 20-22 age bracket. The participant group was academically diverse, representing a range of majors including Cognitive Science, Environmental Science, Education, Nanoengineering, and Animal Science.

5.2 Background Information

All interview participants actively use various planning methods for their academic tasks. Six participants utilized a combination of digital and physical planning tools. A preference for a combination

of digital and physical planners (6) was more common than the exclusive use of physical planners (3) and digital planners (1). Digital tools varied: three students used Google functions, including Google Calendar (all 3), Gmail (1), and Google Spreadsheet (1). Apple's built-in functions, such as Apple Calendar (2) and Apple Reminder (2), were also popular. Additionally, two participants used Canvas Calendar, and one employed Excel. Those using physical tools often did so across multiple devices. Many indicated a desire for improved features in digital planning apps and experienced varying degrees of stress relief after organizing their tasks.

5.3 Set up and Method

These one-on-one follow-up interviews revolved around semi-structured open-ended questions aimed at exploring participants' current planner use, their motivations, and the features they find most useful or lacking in their planning tools. All interviews and observations were recorded and transcribed. Digital planning products were saved via screenshots, while physical planners or mock planning papers were scanned. Interviewers then summarized details of each participant's planning habits. We applied Cross-Analysis to assess each question individually, identifying commonalities and differences across responses from respondents with digital, physical, and both digital and physical planners. Subsequently, we conducted a collaborative Thematic Analysis to draw broader themes and patterns from the dataset. Recognizing potential researcher biases, all four researchers were involved to ensure inter-reliability and reduce subjectivity in interpreting the data.

6 INTERVIEW AND OBSERVATION RESULTS

We structured the results of our interviews and observations into higher-level themes for clarity. We have summarized the key reasons provided by college students for their continued use of physical planners, and for those who exclusively use physical planners, why they prefer them over digital options. Additionally, we explored the perspectives of students who utilize both physical and digital planners, focusing specifically on what they perceive to be lacking in their digital planning systems.

6.1 Reasons on Physical Planner Usage

6.1.1 Pre-existing Habits

Three participants reported that their use of physical planners began early in their academic journey. One started using a planner in elementary school and continued this practice through middle and high school, while the other two began with academic planners provided by their high school. They find it effective, natural, and logical to continue this habit.

6.1.2 Tangibility and Tactile Engagement

Among the 9 participants who reported using physical planners, 8 emphasized the importance of its tangible nature. They noted that physically writing things down aids memory retention, as they feel more engaged and better remember and understand their tasks. Participants expressed that they tend to forget tasks more easily when they are not written down. Furthermore, the simple action of crossing or checking tasks off in a physical planner was noted to significantly boost motivation. Participants described this process as enjoyable and engaging, providing a sense of fulfillment and accomplishment. Finally, the act of physically holding and interacting with a planner was described as adding a sentimental and physical dimension to the planning experience. Participants appreciated the tangible aspect of flipping through the pages, which allowed them to physically feel the passage of time and the volume of tasks. They also noted an emotional attachment to the planner, which holds their daily thoughts and reflects personal nuances, such as the mood of their handwriting, the presence of doodles, or the use of stickers.

6.1.3 Aesthetics and Creative Expression

Physical planners offer unique opportunities for self-expression, contributing to a sense of personal identity. Participants have indicated that they enjoy the freedom to doodle, skip through, and reuse components, and even tear out pages – a level of flexibility not typically found in digital planners. This sense of ownership and personalization leads to a feeling of completion, and sometimes leads to a sense of personal achievement. Furthermore, physical planners provide participants with the ability to tailor their templates according to individual needs effortlessly and without the need for specialized knowledge often required in digital planning systems. This offers a more interactive and personalized planning experience, accommodating diverse preferences and organizational styles and allowing for a more intuitive and user-friendly approach to planning. Finally, our findings show that 9 out of 10 participants report that the ability to color-code, personalize, and interact with planners through elements like clean layouts, attractive covers, and customizable templates significantly enhances their planning activities. These design elements not only improve mood but also enhance functionality by allowing better organization and categorization. This in turn encourages regular use and updating of planners. One participant emphasized the importance of personal expression, remarking, 'I am a big sticker person, so stickers would be awesome.'

6.1.4 Relaxation and Reflection

Participants reported using digital tools primarily for task-related and mandatory activities like meetings, lectures, and deadlines, which are more about organizing mandatory tasks, both long-term and short-term. In contrast, physical planners are used for more reflective and aspirational purposes, such as making long-term plans and setting future goals. Moreover, participants experience a sense of lowered stress when engaging with physical planners, particularly when reflecting on past achievements and thoughts recorded in their writings. Writing down thoughts provides emotional catharsis, contributing to

stress relief. One participant described this process as a “Brain Dump,” using blank pages in their journal to release thoughts and clear mental space. This practice, along with the satisfaction of checking off completed items, offers a therapeutic and relaxing experience.

6.1.5 Convenience for Low Digital Proficiency

For individuals with limited access to devices, less exposure, or lower proficiency with digital tools, physical planners offer a more convenient alternative. Carrying a small, physical planner and jotting down details can be simpler and more straightforward than navigating electronic devices to mark locations, durations, and other details. Additionally, the minimal effort required to write and draw on paper is a significant advantage, especially for those who are more familiar with traditional methods of planning and note-taking.

6.1.6 Avoiding Distractions with Physical Planners

One participant specifically mentioned their preference for using physical planners as a strategy to limit distractions from digital devices and electronics. By opting for a physical planner, they effectively reduce the potential interruptions and focus-diverting temptations often associated with electronic devices, thereby fostering a more concentrated and undistracted planning environment.

6.2 Major Advantages and Benefits of Digital Planning

6.2.1 Convenience

Participants emphasized the convenience of digital planners, particularly their ability to sync across devices, making them accessible wherever and whenever needed.

6.2.2 Digital Nature

Participants appreciated the digital nature of planning, especially in an academic context where most work is digital. "Since all my university work is digital, using a digital planner just makes more sense," stated a participant.

6.2.3 Reminders

Interviewees highlighted the usefulness of reminders for tasks, due dates, and online meetings, which helped them stay organized. "The reminders keep me on track, especially for online classes and assignment deadlines," mentioned another participant.

6.2.4 Low Maintenance

Participants noted that digital planners come pre-installed on devices, are easily accessible, and are usually free, making them low maintenance.

6.2.5 Paperless

Digital planning is environmentally friendly and space-efficient as it eliminates the need for physical paper and reduces clutter since all information is stored digitally.

6.3 Areas for Improvement in Digital Planner Design

6.3.1 Flexibility and Customization

Users expressed a need for greater variability and flexibility in digital planner design. They reported feeling constrained by rules and limitations imposed by some digital planning tools. One participant specifically mentioned an issue with Canvas not displaying partial completion of assignments, suggesting the need for more nuanced progress-tracking features.

6.3.2 Streamlined Reminder Setup

Interviewees highlighted the importance of a shortcut for setting reminders in digital planners, especially in applications like Apple Calendar. Participants also mentioned the need for the ability to set reminders for future tasks or events efficiently.

6.3.3 Distinct Boxes or Checklists

Users expressed a desire for clearer organization within digital planners, emphasizing the importance of distinct boxes or checkboxes for tasks and events.

6.3.4 Desktop Widget for Task Input

Participants suggested the inclusion of a widget on the desktop for quick and easy input of tasks and events. Such a widget could help bridge the gap between physical and digital planning and facilitate the transition for users.

6.3.5 Improved Task and Calendar Sync

Users expressed the importance of seamless synchronization between reminders and calendars, particularly in Apple's ecosystem. Ensuring that reminders and calendar events are synchronized accurately is crucial for effective planning.

6.3.6 Enhanced Task Visibility

Participants desired the ability to see tasks for the day at a glance within their digital planners. A feature that highlights or summarizes daily tasks would aid in better task management.

6.3.7 Task-Work Time Blocking and Completion

Users found it challenging to block out specific task-work time and check off completed tasks within the digital planner.

6.3.8 User-Friendly Learning Curve

Participants noted that adapting to a new planning system often requires a learning curve. Design should aim for user-friendliness and ease of adoption to minimize initial challenges.

6.3.9 Improved Assignment Management

Digital planners should offer a more efficient way to manage assignments and tasks, especially in an educational context. Addressing issues like Google Calendar's inability to display a to-do list and Canvas Calendar showing all assignments can enhance the user experience.

6.3.10 Simplified Color-Coding

While color-coding is a valuable feature, it should be made more intuitive and flexible. Users should have the ability to color-code without affecting past entries, providing a more user-friendly experience.

7 DISCUSSION AND DESIGN IMPLICATIONS

From the results of our survey, interviews, and observations, it is evident that participants employ various tools for effective management of their academic work, with preferences varying due to different reasons outlined in previous sections. Many participants use a combination of both physical and digital planners, seeking to harness the digital convenience of electronic tools alongside the more sentimental and holistic experience offered by physical planners. This section delves into the design implications arising from our comprehensive research. We aim to analyze how these insights can guide the development and enhancement of digital planning tools, ensuring they cater effectively to the diverse needs and preferences of users.

Our study's demographic range in age, gender, and majors revealed a potential bias in our snowball sampling method, as most survey participants were STEM majors. Among interviewees, who varied from STEM to Arts, the majority were female, with a small portion of non-binary and no male representation. This demographic skew might suggest self-selection bias. Additionally, the small sample size due to time constraints means our results might not fully represent the broader college population. These limitations highlight the need for further research with larger, more diverse samples in future studies.

7.1 Focusing on Tangibility and Tactile Engagement

The tactile nature of physical planners plays a significant role in user engagement and memory retention. Designers of digital planners might consider how to incorporate tactile elements or feedback to mimic the physical experience of writing. The action of crossing off tasks in a physical planner is both motivating and satisfying. Digital planners could replicate this feeling by creating more engaging and interactive task completion features, such as page-flipping animations or tactile feedback on touchscreens. This could help in recreating the emotional connection users feel with physical planners.

7.2 Enhancing Customizability and Personalization in Digital Tools

To capture the personal nuances that physical planners offer, digital systems could include more advanced customization options. This could involve allowing users to change backgrounds, themes, or even the layout of the planner to mirror their personal style, similar to using stickers or doodling in a physical planner. Implementing features for personal annotations, like digital stickers or handwriting options, can help users personalize their digital planner in a way that resonates with them emotionally.

7.3 Enhancing Ease of Use and Accessibility

For both digital and physical planners, simplifying the process of task entry is crucial. Users value the immediacy of writing tasks in physical planners, suggesting that digital planners could benefit from reducing the steps required for inputting tasks. Incorporating more persistent properties in features such as reminders in digital planners can address the common issue of forgetting tasks. This could be particularly beneficial for users accustomed to the tangible reminders of physical planners.

7.4 Facilitating Emotional Engagement

Features that enable users to reflect on their achievements or revisit past entries can enhance the sentimental value of a digital planner. This might include 'memory lane' features or summaries of past accomplishments. Implementing mood tracking or journaling features within the planner can help users express and track their emotional states over time, adding a personal and emotional layer to the planning process.

7.5 Improving Assignment Management

For academic or work settings, managing assignments efficiently is key. Digital planners could incorporate features for tracking assignment progress, integrating with educational platforms, and allowing for easy update and review of tasks. For example, adding some features to show progress or mark partially completed assignments would be beneficial.

8 CONCLUSION

In conclusion, our study reveals a combined preference for physical and digital planners among college students. Through surveys, interviews, and observations, we identified the emotional and practical reasons for this choice. While physical planners provide emotional connection and personalization, digital planners offer functional benefits but lack in personal touch. Our findings indicate that many students prefer a blend of both, combining the convenience of digital tools with the sentimental experience of physical planners. These insights pave the way for enhancing digital planner design, aiming to meet the diverse needs and preferences of users in academic settings.

REFERENCES

- [1] Alexander Wilson and Mark Tewdwr-Jones. 2022. Covid-19 and the rise of digital planning: Fast and slow adoption of a digital planning system. *Town Planning Review* 93, 5 (January 2022), 495–518. <http://dx.doi.org/10.3828/tpr.2022.3>
- [2] Amid Ayobi, Paul Marshall, and Anna L. Cox. 2020. Trackly. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. <http://doi.org/10.1145/3313831.3376809>
- [3] Amid Ayobi, Tobias Sonne, Paul Marshall, and Anna L. Cox. 2018. Flexible and Mindful Self-Tracking: Design Implications from Paper Bullet Journals. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18). Association for Computing Machinery, New York, NY, USA, Paper 28, 1–14. <https://doi.org/10.1145/3173574.3173602>
- [4] Keita Umejima, Takuya Ibaraki, Takahiro Yamazaki, and Kuniyoshi L. Sakai. 2021. Paper Notebooks vs. mobile devices: Brain activation differences during memory retrieval. *Frontiers in Behavioral Neuroscience* 15. <http://doi.org/10.3389/fnbeh.2021.634158>
- [5] Mohamad Zukriyani, F.A. and Azizan, N. 2023. Student academic planner system: A Review. *Malaysian Journal of Science Health & Technology*. 9, 1 (2023), 63–73.
- [6] Parastoo Abtahi, Victoria Ding, Anna C. Yang, Tommy Bruzzese, Alyssa B. Romanos, Elizabeth L. Murnane, Sean Follmer, and James A. Landay. 2020. Understanding Physical Practices and the Role of Technology in Manual Self-Tracking. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 4, Article 115 (December 2020), 24 pages. <https://doi.org/10.1145/3432236>
- [7] Robertson, G., Horvitz, E., Czerwinski, M., Baudisch, P., Hutchings, D.R., Meyers, B., Robbins, D. and Smith, G. 2004. Scalable fabric. *Proceedings of the working conference on Advanced visual interfaces*. (2004).
- [8] Ryder Carrol. 2015. *How to Bullet Journal*. YouTube. Retrieved December 10, 2023 from <https://www.youtube.com/watch?v=fm15cmYU0IM>
- [9] Velikova G, Wright EP, Smith AB, Cull A, Gould A, Forman D, Perren T, Stead M, Brown J, Selby PJ. 1999, Automated collection of quality-of-life data: a comparison of paper and computer touch-screen questionnaires. *Journal of clinical oncology*, 17(3) 998-998. <https://doi.org/10.1200/jco.1999.17.3.998>
- [10] William Jones, Predrag Klasnja, Andrea Civan, and Michael L. Adcock. 2008. The personal project planner: planning to organize personal information. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08). Association for Computing Machinery, New York, NY, USA, 681–684. <https://doi.org/10.1145/1357054.1357162>
- [11] Yanliu Huang, Zhen Yang, and Vicki G. Morwitz. 2020. How using a paper versus mobile calendar influences everyday planning and plan fulfillment. *Journal of Consumer Psychology* 33, 1, 115-122. <https://doi.org/10.1002/jcpy.1297>
- [12] Zhen Li, David Tinapple, and Hari Sundaram. 2012. Visual planner: beyond prerequisites, designing an interactive course planner for a 21st century flexible curriculum. In CHI '12 Extended Abstracts on Human Factors in Computing Systems (CHI EA '12). Association for Computing Machinery, New York, NY, USA, 1613–1618. <https://doi.org/10.1145/2212776.2223681>

APPENDIX

A. Qualitative Survey Question Questions, Themes, and Graphs

Q1: What factors (e.g. convenience, accessibility, customizability, habit, recommendation of others, etc.) influenced your current choice of planning system?

Themes: Accessibility/Convenience, Customizability, Informative View, Previous Experience/Habit, Ease of Use, Integration with Other Apps, Digital Features, Physical Presence, Physical Writing, Simplicity/Learnability, and Other.

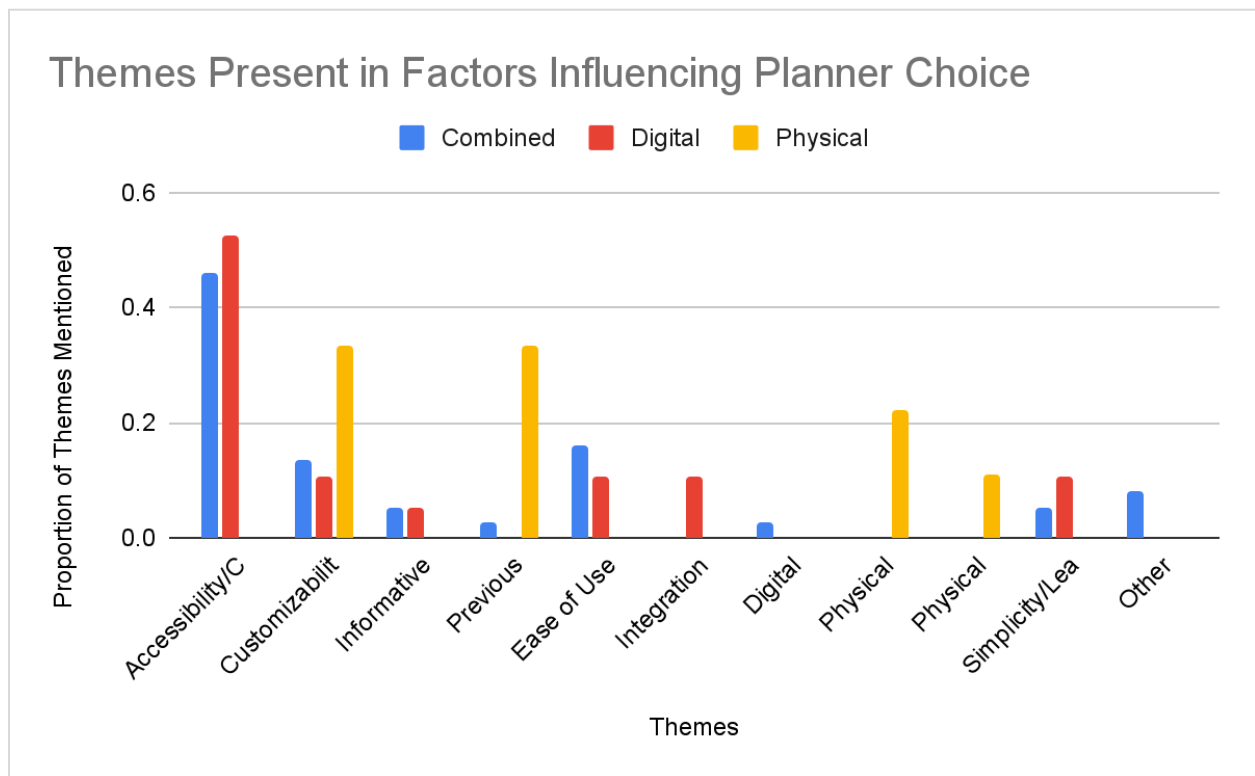


Figure 1: Proportion of Themes Mentioned in Response to Question 1.

Q2: What aspects of your current planning system do you appreciate most?

Themes: Accessibility/Convenience, Customizability, Informative View, Previous Experience/Habit, Ease of Use, Flexibility, Integration with Other Apps, Digital Features, Physical Presence, Physical Writing, Organization/Structure, Other.

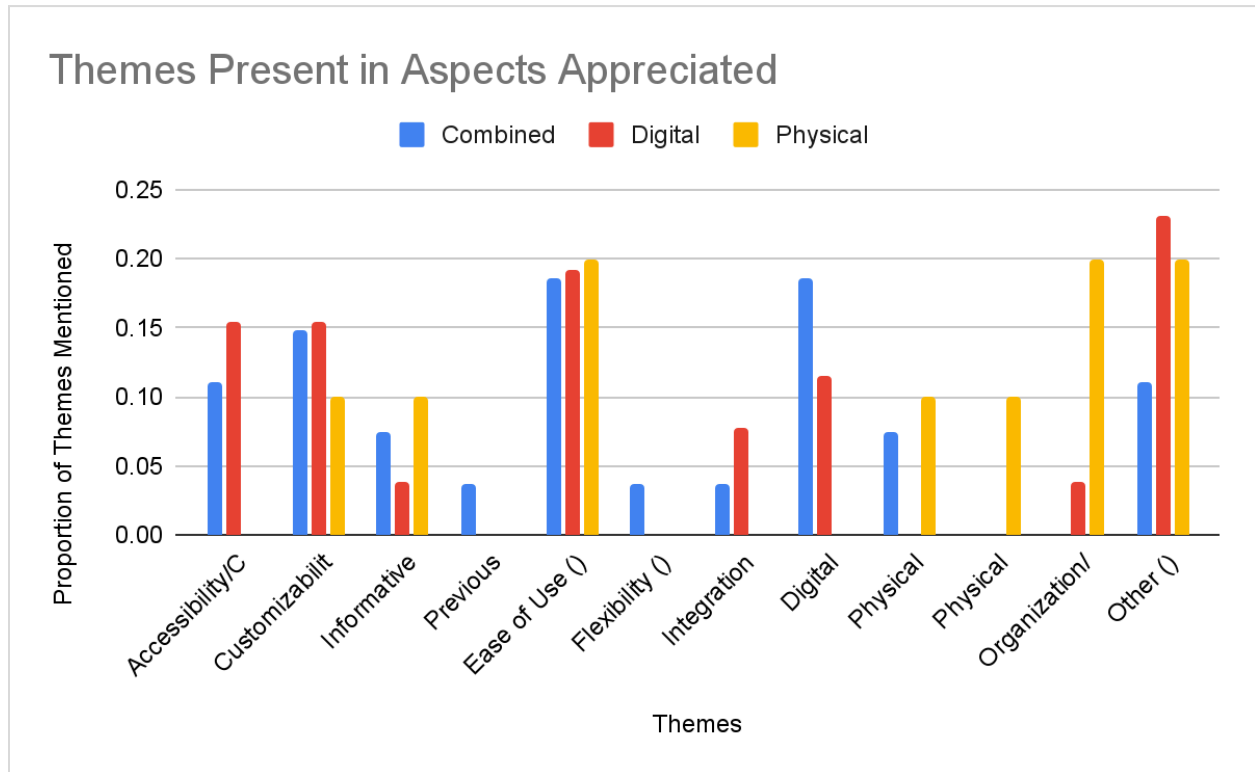


Figure 2: Proportion of Themes Mentioned in Response to Question 2.

Q3: What problems or incidents have you encountered when planning academic tasks with your current system?

Themes: Difficulty Planning, Lack of Presence, Unclear/Difficult Controls, Difficulty Modifying Tasks, Inaccessibility, Inflexibility, Unsynchronized Planners, Uninformative/Suboptimal View, Tedious to Use, User Habits/Consequences.

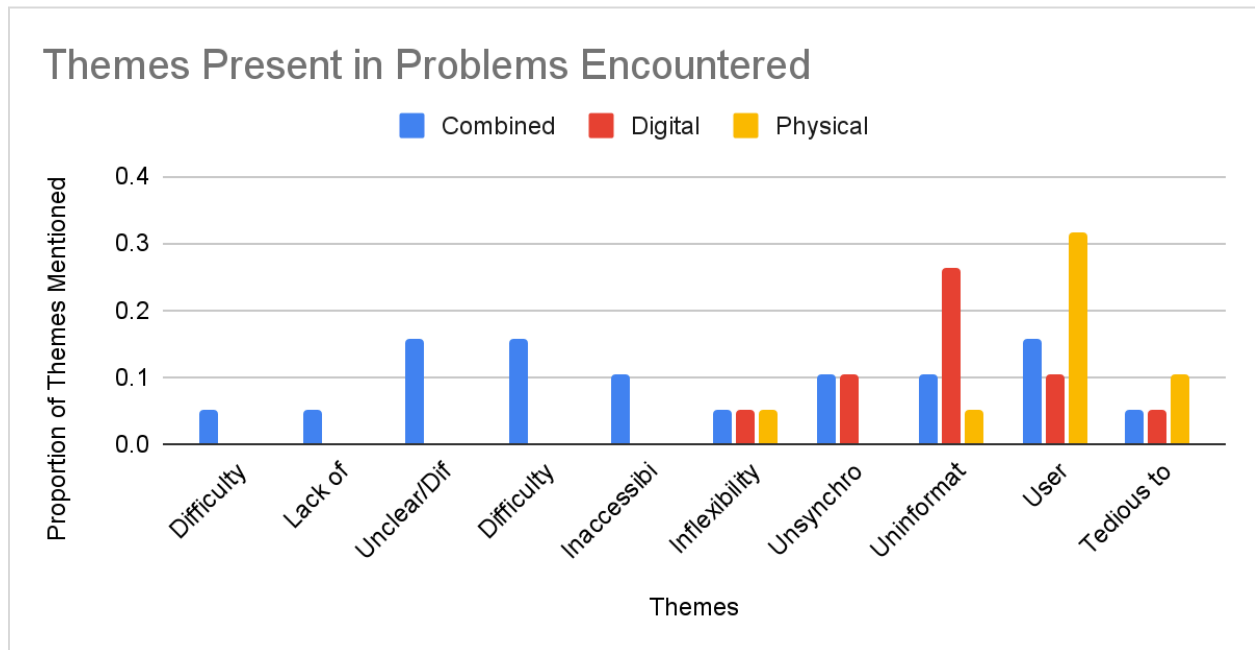


Figure 3: Proportion of Themes Mentioned in Response to Question 3.

Q4: In your opinion, what is missing or could be improved in your current planning system?

Themes: Extra Section, Roomier Layout, Better Organization, More Task Information, Improved View, Synchronization Across Devices/Applications, Better Controls, Reminders, Better Customization, Other.

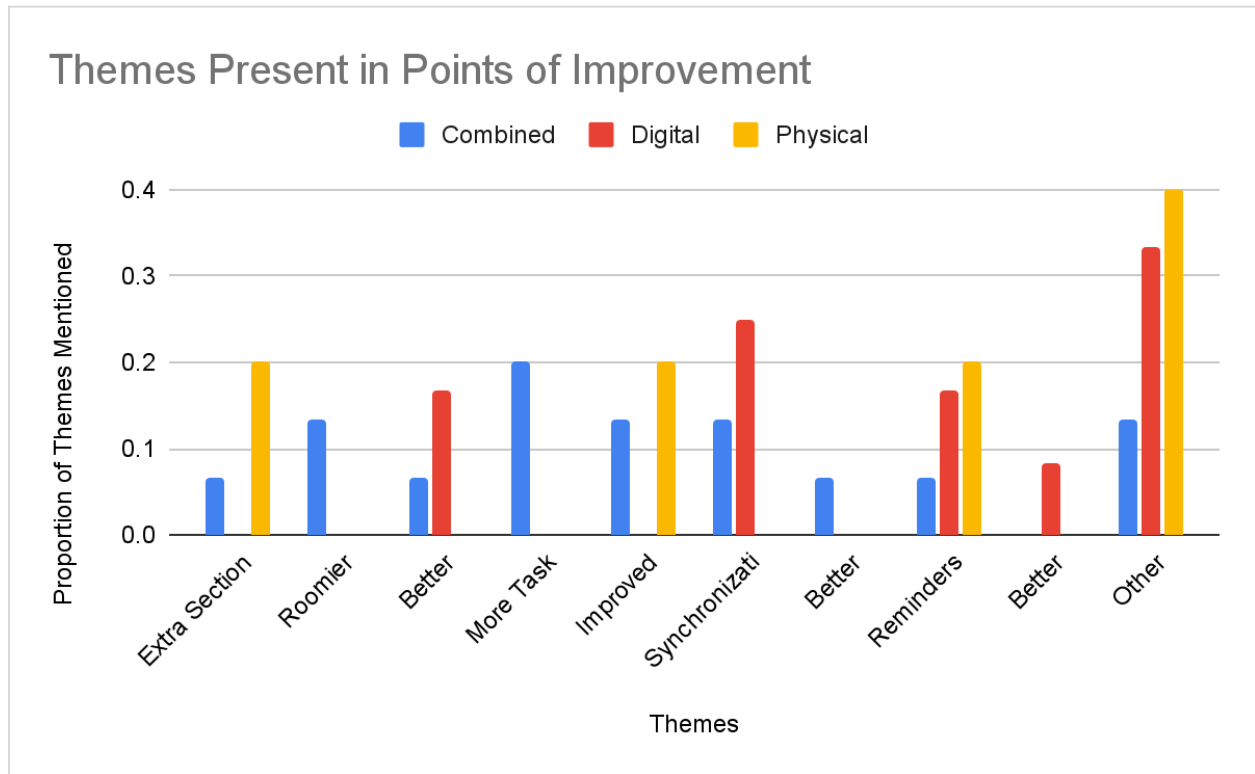


Figure 4: Proportion of Themes Mentioned in Response to Question 4.

Q5: Are there any cases where you prefer one kind of planner over the other? Please elaborate.

Themes: Short-Term/Day-to-Day Tasks, Long Term/Large Tasks, Initially Write Tasks, Track/View Tasks, Academic Tasks, Personal Tasks, Other.

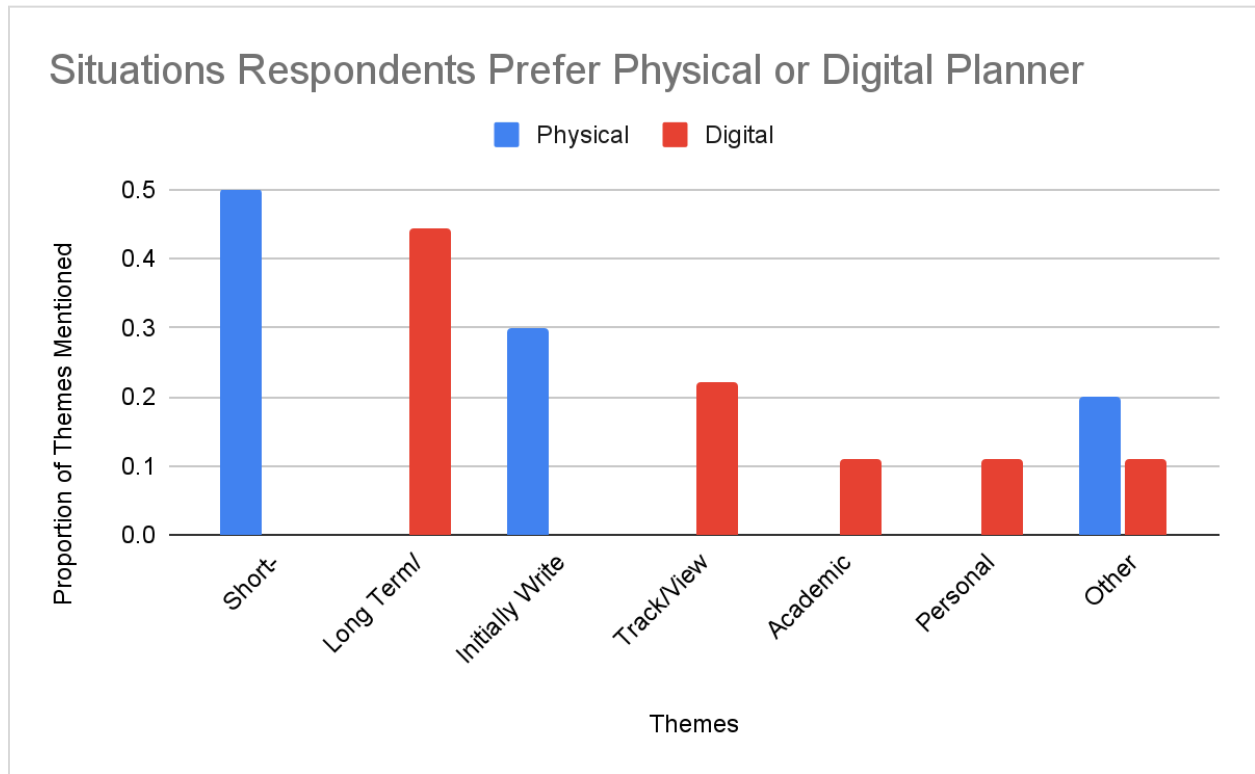


Figure 5: Proportion of Themes Mentioned in Response to Question 4.

B. Sample Interview Questions and Observations

Informed consent: This interview will take around 20 minutes. I will keep the information you give me confidential; in no way will the others be able to identify that it was you giving these particular answers.

1. Would it be OK if I record this session? This is for documentation purposes only. Do you have any questions for me before we begin?

START RECORDING NOW WITH ZOOM VIDEO OR AUDIO

Questions:

Current Planner Use

Can you walk me through how you typically use your planner on a daily basis?

- Follow-up: What features do you use the most?

What motivated you to start using your current planning system?

- Follow-up: Have these motivations changed over time?

How do you decide between using a digital or physical planner?

- Follow-up: Are there emotional factors (elaborate) that influence this choice?

Features and Preferences

What features do you find most useful in your current planner?

- Follow-up: Are there features you wish were improved or added?

Can you describe a time when your planner was particularly helpful in managing your tasks or schedule?

- Follow-up: Was there a time when it fell short of your needs?

How do design elements (like color, layout, interactivity) affect your planning experience?

Emotional and Behavioral Aspects

Do you feel your planner reflects your personal style or identity? How so?

How does planning and organizing your tasks impact your stress levels or overall well-being?

Could you share any specific frustrations or challenges you've experienced with your planning system?

Task for Observation:

Structured observation (Say tasks step by step, expand and make tasks messier):

Imagine you are in class on Monday. Your professor assigns an essay that you estimate will take about 5 hours, due at 11:59 pm on Saturday. In addition, you remember that you still need to plan a group meeting to finish a homework assignment, due at 11:59 pm on Friday. Later, on Canvas, you see a reading response due at 12 pm on Saturday and you haven't touched the reading yet. Please use your current system to input these tasks into the planner.

Please input any of the following tasks you would normally put into your planner in your planner. If you would normally include more details, ask what those details are. Pretend it is Monday, December 4.

***Introduce main bullet points, subpoints are in case they ask for more details (just update subpoints if they ask for info that isn't there). Take note of what information they needed and how they included tasks.**

- Your professor assigns a large essay due 11:59 PM Saturday, December 9
 - will take about 5 hours
 - due 11:59 PM
 - class is MOD 130
 - essay topic is "Development of Modern Healthcare"
- You need to plan a group meeting to finish a homework assignment due 11:59 PM on Friday, December 8
 - Expect meeting to take 2 hours
 - class is MOD 120
- You see that a reading response is due at 12 PM on Friday, December 8
 - haven't started reading
 - class is MOD 120
 - expect it to take 2 hours
- There is a quiz you haven't studied for at 12 PM on Friday, December 8
 - class is MOD 120
 - expect to study for 3 hours
 - worth 10% of grade
- Your professor introduces a multi-step final project due on December 21 at 11:59 PM
 - class is MOD 140
 - 3 phases
 - planning phase due December 12
 - project execution due December 16
 - project presentation due December 21
 - group project

After completing the task, you can ask reflective questions to gain deeper insights:

- What was your thought process while completing this planning task?
- Did you face any difficulties or uncertainties during the task?
- How representative is this task of your typical planning routine?
- For a certain task: Why did or didn't you include this task?